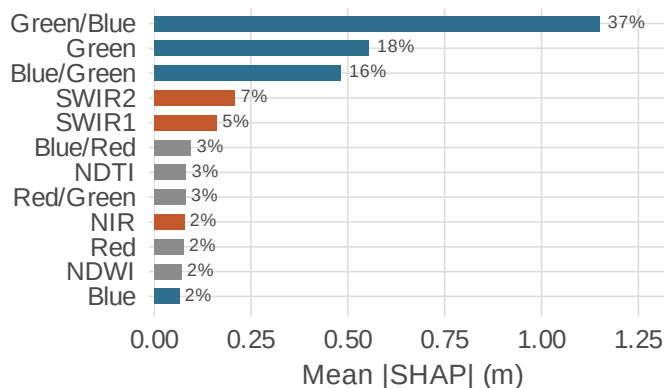


(a) Green and blue dominate attribution

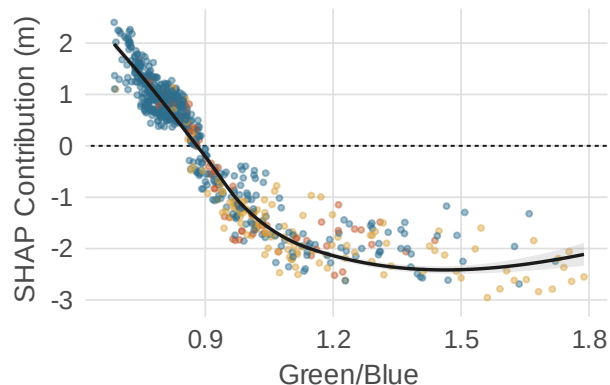
Mean |SHAP| computed OUT-OF-FOLD: each observation is explained by the model that did not see it. Labels are the share of the total



■ Green/blue ■ NIR/SWIR ■ Other

(b) How Green/Blue acts

SHAP value vs. feature value. Greener water relative to blue pushes the prediction toward less transparent water. Trimmed to the 1-99th percentile (18 extreme points out)



● Bahía de Puno ● Lago Mayor
● Lago Menor

(c) Only transparency is retrievable in this water

Random Forest with the same honest design (campaign blocking) for each variable.

The negative claim is subject to the same rigor as the positive one

